Assignment 1

Hotel Management System

Course: CCAS 4.3-2

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Functional Requirements:

ID	Requirement	MoSCoW	Justification
FuncReq-1	The database shall record the payment status, such as unpaid, deposit or paid.	**Must Have**	Staff need to know whether a guest's bill has been settled.
FuncReq-2	The database shall store payment records, including payment ID, reservation, amount, date, and payment method.	**Must Have**	Payment records help the hotel track money received.
FuncReq-3	The database shall record the actual check-in date and check-out date , time for each guest stay exactly.	**Must Have**	This records the guest's check-in and check-out dates and times, identifying when their stay begins and when the room becomes available again after departure and cleaning.
FuncReq-4	The database shall prevent overlapping confirmed reservations for the same room and dates.	**Must Have**	A room cannot be assigned to two guests for overlapping stays.
FuncReq-5	The database shall store each guest's ID, full name, phone number, email, and address.	**Must Have**	Guest information is necessary to manage reservations ,safety and hotel stays.

FuncReq-6	The database shall allow authorized staff to view and update guest information.	**Must Have**	Guest details may change and must remain accurate.
FuncReq-7	The database shall assign a unique ID to every hotel room.	**Must Have**	Each room must be identified individually.
FuncReq-8	The database shall store room types, such as single, double, suite, and family room.	**Must Have**	Room types help staff match accommodation to guest needs.
FuncReq-9	The database shall link every reservation to the guest who made it.	**Must Have**	The hotel must identify the person responsible for each booking.
FuncReq-10	The database shall allow staff to check room availability for a requested date range.	**Must Have**	This helps prevent booking rooms that are already reserved.
FuncReq-11	The database shall allow authorized staff to create a reservation.	**Must Have**	Creating bookings is a core hotel operation.
FuncReq-12	The database shall allow authorized staff to modify reservation dates or assigned rooms.	**Must Have**	Guests may change their travel plans,and thats their whole right.
FuncReq-13	The database shall record reservation cancellation dates and statuses.	**Must Have**	Staff need to know which bookings are no longer active.
FuncReq-14	The database shall record additional charges, such as massage,restaurant bills, laundry,drinks or room service, against the relevant guest stay.	**Should Have**	Guests may use services beyond the room itself.
FuncReq-15	The database shall store staff details,	**Must Have**	Staff records support hotel operations and

	including staff ID, name, job role, and contact information.		accountability.Also for extra safety.
FuncReq-16	The database shall maintain a record of guest complaints and their resolution status.	**Could Have**	Complaint records can help improve service quality.
FuncReq-17	The database could store guest loyalty points and membership levels.	**Could Have**	A loyalty feature may encourage repeat bookings but is not essential to the initial system.
FuncReq-18	The database shall provide reports showing current occupancy and room availability.	**Should Have**	Management needs to monitor how hotel rooms are being used.
FuncReq-19	The database shall allow staff to mark a room as unavailable during maintenance.	**Must Have**	A room undergoing maintenance should not be offered to guests.
FuncReq-20	The database shall calculate or store the total room charges based on the applicable nightly rate and billable nights.	**Must Have**	The hotel needs accurate accommodation charges.

Non-Functional:

ID	Requirement	MoSCoW	Justification
NonF-1	The system shall require users to authenticate before accessing protected functions.	**Must Have**	Authentication prevents unauthorized access to hotel records.
NonF-2	The database shall restrict access to guest, payment, and staff information according to user roles.	**Must Have**	Different staff members should only access information necessary for their duties.
NonF-3	The system shall protect sensitive information during transmission using secure communication protocols.	**Must Have**	This reduces the risk of personal information being intercepted.
NonF-4	The system shall complete a room reservation transaction within 3 seconds under normal operating conditions.	**Must Have**	Fast processing improves the booking experience and helps staff handle reservations efficiently.
NonF-5	The database shall be backed up regularly, at least once every 24 hours.	**Must Have**	Regular backups help recover information after data loss or system failure.
NonF-6	The system could support multiple languages in its interface and guest-facing booking information.	**Could Have**	Language options can improve accessibility for international guests.

NonF-7	The system shall maintain consistent date and time formats across reservation and check-in/check-out records.	**Must Have**	Consistent timestamps help avoid confusion about booking periods and guest stays.
NonF-8	The database design shall support an increase in rooms, guests, reservations, and transactions without requiring a complete redesign.	**Should Have**	The hotel may expand its operations in the future.
NonF-9	The system shall record important changes to reservations, payments, and room statuses with the responsible user and timestamp.	**Should Have**	Audit records help investigate errors and unauthorized changes.
NonF-10	The system shall validate entered data before saving it to the database.	**Must Have**	Validation reduces errors such as invalid dates, negative prices, and missing information.